BUSINESS PROPOSAL

Flow Management and Transshipment Scheduling Platform

Prepared for PT Asian Bulk Logistics (ABL)

Prepared by Vector Consulting Indonesia for Berau Coal transshipment operations in East Kalimantan. This proposal sets out a flow-management approach for synchronising shipment commitments with ABL port, blending, barging, transshipment, survey and OGV loading execution.

Proposal item	Detail
Purpose	Propose a Flow Management-led scheduling and control-tower approach for Berau-ABL operations.
Source basis	ABL company profile, Scheduling Simulation BRD, ABL scheduling discussion transcript and proposal-context synthesis.
Delivery focus	Diagnostic, solution blueprint, platform build, testing, training, go-live support and operating handover.
Date	May 2026

60+	**2,000 tph**	**55,000 t/day**	**20M+ tpa**	**55-60%**
Barges managed	BLC capacity	Peak CTS capacity	Blending capacity	Indicative fleet productive time

Proposal lens. Berau defines what must be shipped, when, to which buyer and at what quality. ABL must convert that demand into an executable flow through port, blending, tug-barge, CTS and OGV resources.

PURPOSE AND BUSINESS CASE

1. Executive Summary

PT Asian Bulk Logistics (ABL) operates a deeply interdependent coal logistics chain in East Kalimantan, converting Berau Coal's monthly and vessel-wise shipment commitments into executable port, blending, tug-barge, transshipment and OGV loading operations.

The business problem is not only fleet allocation. It is a full flow-management problem across dynamic coal quality decisions, multiple jetties with very different cycle times, tide and bridge windows, CTS availability, survey milestones and OGV laycan commitments. A plan that is locally correct at the jetty or vessel level can still fail if the full chain is not synchronised.

The current working reality is still heavily dependent on Excel-based schedule revisions, WhatsApp or voice coordination and partial visibility from AIS/Spinergie. AIS can show where an asset is, but it does not reliably confirm business state: loading, discharging, waiting for tide, grounded, waiting for quality release, waiting for survey, reassigned or truly available.

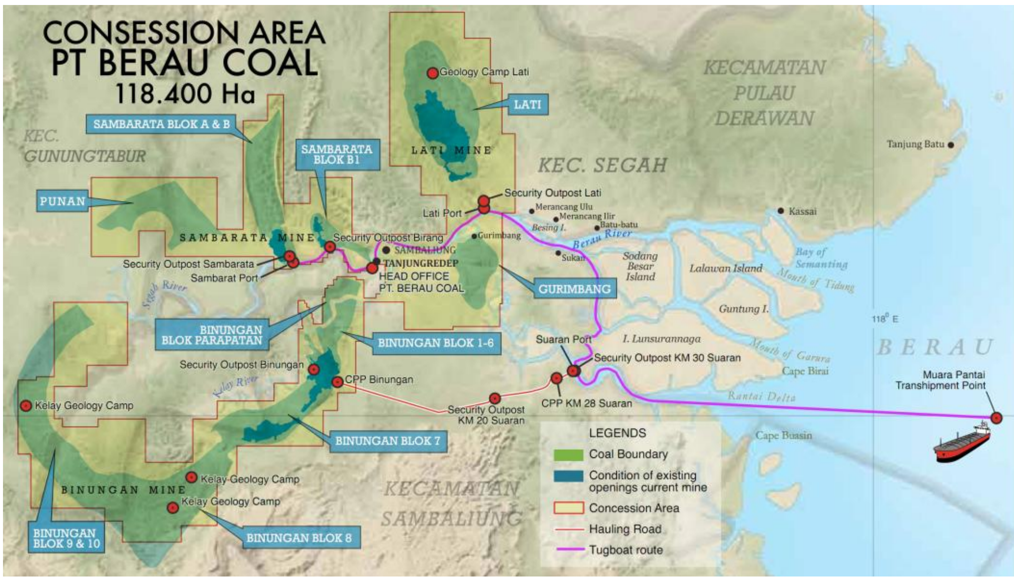
Vector Consulting Indonesia proposes a Flow Management and Transshipment Scheduling Platform: a governed control-tower layer that integrates shipment demand, cargo and blend readiness, port and BLC capacity, tug-barge assignments, tide and bridge feasibility, CTS/OGV loading, AIS actuals and exception recovery into one ABL-governed planning and decision layer, using Berau demand inputs and agreed handoff outputs.

BUSINESS CONTEXT

2. ABL, Berau Coal and Operating Context

ABL is positioned as an integrated logistics and infrastructure solutions provider. For the Berau operating context relevant to this proposal, the scope is limited to port and blending readiness, BLC operations, tug-barge execution, CTS/transshipment coordination, survey milestones and OGV loading handoff.

Berau Coal is the cargo owner and demand generator in this operating chain. Commercial demand is translated into shipment programmes, customer-specific coal quality requirements, vessel nominations, laycan windows and OGV loading expectations. The operational demand ABL must execute is therefore a time-phased cargo movement requirement, not a simple transport order.



Berau corridor from mine and port sources through river route to transshipment point.

Relationship and KPI lens. The solution must support a shared view of customer delivery, laycan adherence, demurrage avoidance, flow time and asset productivity across Berau and ABL.

FROM DEMAND TO OGV DEPARTURE

3. End-to-End Operating Scope

The proposed platform scope covers the full physical and information flow from shipment demand to OGV departure with loaded cargo. The scope is intentionally wider than fleet scheduling because the fleet plan is only feasible when cargo readiness, port readiness and marine constraints are aligned.

END-TO-END OPERATIONAL FLOW — FROM SHIPMENT COMMITMENT TO OGV DEPARTURE

STAGE 1 Mine / CPP Stockpile	STAGE 2 Jetty / BLC Loading	STAGE 3 Tug + Barge River Run	STAGE 4 Tide · Bridge Windows	STAGE 5 CTS / FTS Float. Crane	STAGE 6 OGV Loading & Survey	STAGE 7 Customer Delivery

Cross-Cutting Constraints	Demand & OGV Laycan · Coal Grade / Blend Recipe · Quality Release Status · Tide Windows · Bridge Crossing Windows · River Draft / Choke Points · Asset Availability (closed-loop) · Hatch Plan & Layering Sequence · Survey & Certification Milestones · AIS / GPS Actuals · Third-Party Asset Lead Time

Mine / CPP / Stockpile	Jetty / BLC Loading	Tug + Barge / River	Tide · Bridge Windows	CTS / FTS / Float. Crane	OGV Loading & Survey	Customer Delivery
– Grade & spec – Blend recipe – Stockpile qty – Quality release – Ready-to-load – Mine routing	– BLC slot & rate – Jetty queue – Loading window – Conflict flags – Survey readiness – Plan vs actual	– Tug HP + pairing – Load/sail/return – Choke-point check – Breakdown log – Charter tracking – Closed-loop avail	– Tidal calendar – Bridge & draft – Feasibility check – Seasonal restrict. – Missed-window alert – Buffer planning	– Asset-type match – Discharge rate – Barge queue – Downtime & metal – Anchorage zone – CTS vs OGV demand	– ETA / laycan – Hatch & layering – Barge sequence – Draft survey – Completion proj. – Demurrage risk	– Cert of wt/quality – Statement of facts – Shipment priority – Delay attribution – Shipment status closure – Governed handoff

PLAN	SIMULATE	TRACK	ALERT	RECOVER
Demand → feasible fleet assignment	*What-if scenario comparison*	*Live AIS/GPS vs planned*	*Conflict, delay & risk flags*	*Re-sequence & optimise*

Operating control:	Plan = approved intent	Live/AIS = movement evidence	Inferred = event candidate	Confirmed = accepted actual	Recommended = recovery option

The flow makes clear that ABL's scheduling challenge is not asset location — Spinergie already provides AIS visibility. The challenge is synchronizing cargo readiness, asset availability, environmental constraints and OGV demand into a single executable plan and continuously updated operating view. Approved schedule changes remain governed. Each stage is a scheduling dependency for the next: a quality hold at Stage 1 delays BLC loading at Stage 2; a missed tide window at Stage 4 delays CTS throughput at Stage 5; a barge sequence error at Stage 5 violates the hatch plan at Stage 6. The platform must model and govern the full chain, not individual legs in isolation.

EXISTING SIGNALS AND GAPS

4. Current System and Data Landscape

The present landscape contains valuable digital signals, but they are not yet unified into a constraint-aware scheduling engine. The proposed solution should augment the existing landscape rather than blindly replace it.

System / data source	Available value	Gap to be closed
Microsoft Excel	Current manual planning and schedule revision layer.	Static and labour-intensive; unable to quickly evaluate revised plans against current constraints.
Spinergie / AIS	Smart fleet management, operational reports and vessel position visibility.	Strong position signal, but not a full business-state or scheduling optimiser.
GPS / AIS position feeds	Real-time or near-real-time location evidence for tugs, barges, CTS, OGVs and support craft where feeds are available.	Coverage, latency and signal-only visibility must be handled through feed freshness checks, confidence flags, fallback confirmation and geofence/event enrichment.
Field connectivity	Existing connectivity may support field reporting where available.	Solution must tolerate intermittent connectivity through manual entry, delayed sync, and clear timestamp/source capture.
Operational reports / database	Some operational data exists, but field reporting is inconsistent.	Needs standardised event definitions and adoption discipline.
Environmental constraint inputs	Tide, weather and bridge-window data where available.	Use as scheduling constraints through Excel, manual or API inputs confirmed during diagnostic.
Berau inputs	OGV schedule, cargo quantity, coal quality, layering sequence and jetty/source plan.	Current data changes monthly, weekly and daily; change visibility must be governed.

AIS-specific limitation. AIS should be treated as the live movement signal, not the system of operational truth. The new platform must convert AIS movement into business events and reconciled schedule states.

DISCOVERY FINDINGS (ONLINE MEET + QA)

5. Current Business Challenges

The following challenges are framed as operating realities rather than isolated observations. They define the design requirements for a control-tower platform.

#	Challenge	Operational impact
1	Demand and source-jetty volatility	Monthly schedules are revised weekly and daily. Berau may change source jetties because production and blending requirements change, forcing repeated reallocation.
2	Coal quality and blending dependency	One OGV may require coal from multiple jetties or grades. A tug-barge assigned to a buyer or grade cannot always be reallocated.
3	Very unequal cycle times	Nearest cycles may finish in hours; farthest routes can consume 48-70 hours. Generic allocation hides this mismatch.

#	Challenge	Operational impact
4	Low productive utilisation	Field feedback indicates roughly half of fleet time can become unproductive through queueing, waiting, instruction gaps, tide/bridge waits and reassignment delays.
5	Grounding / low-tide risk	Sometimes intentional waiting before departure is the correct decision, but field teams need system-backed justification.
6	Position without activity truth	Teams may know where an asset is but not what it is actually doing because AIS does not confirm business state.
7	Excel-driven rescheduling delay	When a vessel delay, source change or quality issue occurs, recalculation may take hours while assets wait or move on local judgement.
8	Mid-route reassignment and buyer changes	Tug-barge sets can be redirected based on shipper or field decisions, creating hidden queues and accountability issues.
9	Third-party charter lead time	Additional tugs may require about one week for mobilisation and inspection, so recovery cannot assume immediate extra capacity.
10	KPI misalignment across parties	Jetty productivity, asset utilisation, demurrage and delivery reliability are often managed through separate KPI views.
11	Decision fear in the field	A field team may be blamed for waiting even when waiting avoids grounding or waste; simulation and approval workflow must legitimise good decisions.

TARGET STATE

6. Proposed Solution: Flow Management Control Tower

The proposed solution sits above fragmented tools and operating teams. It creates one governed schedule, one operational truth, one exception view and one recovery decision workflow.

Operating principles

- **Synchronise the full chain:** stockpile, blending, jetty, tug-barge, tide/bridge, CTS and OGV must be planned together.
- **Use choke-point release discipline:** do not release cargo or assets into a known downstream blockage simply to avoid local idle time.
- **Make intentional waiting legitimate:** if tide, bridge or grounding simulation shows that waiting now avoids greater loss later, the system should recommend and record it.
- **Protect operating truth:** keep the approved plan, live AIS evidence, inferred event candidates, confirmed actuals and recovery recommendations separate until a planner-approved workflow changes the schedule.
- **Use shared KPIs:** customer delivery, OGV completion, demurrage risk, flow time and asset productivity should override narrow local measures.

Core functional modules

Module	Purpose
Demand and OGV Programme	Import or enter OGV nomination, laycan, quantity, grade, hatch/layering sequence and shipment priority.
Cargo Readiness and Blending	Track stockpile/source availability, blend recipe, quality status, rain/moisture risk, sampling and release-to-load status.
Port / Jetty / BLC Scheduling	Manage jetty queue, BLC slots, loader rates, loading duration and ready-to-load windows.
Tug-Barge Assignment	Assign owned and chartered tug-barge sets using route cycle time, draft, availability, maintenance and next feasible assignment.
Tide / Bridge Feasibility Engine	Validate departure, passage and arrival windows using tide, bridge opening, draft and route constraints.
CTS / OGV Loading Plan	Assign CTS/floating crane capacity, hatch/layering sequence, discharge queue and OGV completion forecast.
AIS-to-Business-State Layer	Convert position and geofence signals into operational events such as arrived, loading, departed, waiting, discharged and available.
Simulation and Recovery	Run what-if scenarios for OGV delay, source change, grounding risk, breakdown, weather delay and quality hold.
Governed Decision Workflow	Recommend, approve, execute and audit changes so field teams are protected by system-backed decisions.
Management KPI Dashboard	Measure planned vs actual, idle reasons, demurrage risk, utilisation, throughput, waiting by control point and delay attribution.

Target state on the ground. A dispatcher should see each OGV as a demand-to-completion flow: required cargo, blend status, assigned barges, route feasibility, CTS queue, hatch progress, survey status and projected completion. When a constraint changes, the system should show affected trips and propose safe recovery options for planner approval.

PHASED DELIVERY

7. Engagement Roadmap

The delivery roadmap creates control and visibility before advanced optimisation is introduced. It follows the proposal structure while explicitly including process blueprinting and adoption governance.

Phase 0: Diagnostic and Flow Blueprint (Weeks 1-6)

- **Deliverable:** Signed blueprint, solution architecture and MVP scope.
- **Scope:** Map current process, decision rights, data availability, fleet/jetty/CTS master data, tide/bridge logic, Berau inputs, KPIs and adoption risks.

Phase 1: Digital Schedule Board and Master Data (Months 1-3)

- **Deliverable:** Governed schedule board and common master data.
- **Scope:** Create master data and schedule board for OGV, cargo, jetty, tug-barge, CTS and survey milestones. Manual inputs allowed; conflict checklist and planned-vs-actual view included.

Phase 2: Simulation and Constraint Feasibility (Months 4-7)

- **Deliverable:** Constraint feasibility and recovery scenario layer.
- **Scope: Add tide/bridge feasibility, route cycle-time logic, draft/tide/route-window risk flags based on agreed operating rules, OGV completion projection, demurrage risk and what-if recovery scenarios.**

Phase 3: Spinergie / AIS Integration (Months 8-12)

- **Deliverable:** Live event ingestion and planned-vs-actual reconciliation.

- **Scope: Ingest available AIS/GPS/geofence events or imports, reconcile planned vs actual, infer candidate business events and trigger live delay alerts. Continue manual confirmation for uncertain states.**

Phase 4: Event-Driven Operations (Year 2 Q1)

- **Deliverable:** Standardised field event capture.
- **Scope:** Capture controlled mobile or web event updates from tugs, jetties, CTS and survey points to reduce reporting gaps.

Phase 5: Optimisation and Control Tower (Year 2 Q2-Q4)

- **Deliverable: Rule/solver-assisted recovery support and control cockpit.**
- **Scope: Add assignment support, recovery recommendations, an ABL control cockpit with Berau-facing inputs/outputs as agreed, and management KPI dashboards.**

IMPLEMENTATION COMMITMENT

8. Committed Deliverables

Workstream	Key deliverables	Acceptance criteria
Diagnostic	Discovery sessions; current Excel/process mapping; constraint inventory; KPI and delay-reason taxonomy; data availability assessment.	Signed diagnostic and flow-map document.
Solution Blueprint	Future-state process; master data model; integration map; role/decision workflow; MVP and phase scope; UX wireframes.	Approved blueprint and architecture.
Platform Build	Schedule board; demand input; cargo/blend readiness; fleet/jetty/CTS planning; simulation; AIS/Spinergie connector subject to confirmed data access; reporting dashboards.	Functional platform deployed to UAT environment.
Testing and UAT	Scenario tests using real cases: source change, OGV delay, grounding risk, charter shortage and quality hold.	UAT sign-off with critical defects closed.
Training and Adoption	Role-based training for planners, dispatchers, field coordinators and management; Berau-facing handoff guidance where included in scope; quick-reference SOPs.	Users trained and process adoption signed off.
Go-Live and Hypercare	Production cutover; control-room support; KPI baseline comparison; issue log; handover to operations and IT.	System live and operationally adopted.

VALUE CASE

9. Expected Business Outcomes

↑ Fleet & CTS Utilisation From ~55–60% toward planned-utilisation targets through improved synchronisation and queue reduction.	**↓ Avoidable Waiting Time** Measurable reduction in idle time at jetties, CTS, tide waits and OGV holds through constraint-aware scheduling.
✓ OGV Laycan Compliance Hatch and layering plan enforced in the schedule — demurrage risk visible before the vessel arrives.	**⊙ One Shared Schedule** Single governed plan reducing reliance on parallel Excel files, WhatsApp coordination and informal reporting channels.
◎ Cargo & Blend Visibility Grade readiness, blend status and quality release integrated as scheduling inputs, not post-planning discoveries.	**Faster Exception Response** Alerts and scenario comparison aim to reduce recovery decision time through earlier visibility and structured options.
▦ Clear Delay Attribution Delay attribution by control point, reason code and agreed owner/source where available — enabling accountability and improvement.	**◈ Transition to Flow Management** From reactive Excel-and-WhatsApp coordination to a constraint-aware, continuously updated flow management layer.

WORKING MODEL

10. Engagement Approach and Governance

Vector Consulting Indonesia will work with ABL as a combined process-and-technology delivery partner. The project should not be treated as a pure software build. It requires blueprinting, shared KPI design, decision-right governance, field adoption and phased technology enablement.

Governance cadence

- Weekly working sessions during diagnostic and blueprinting.
- Bi-weekly steering review with ABL leadership, operations and IT stakeholders.
- Milestone sign-off at Diagnostic, Blueprint, UAT and Go-Live gates.
- Change-control process for scope, data source, algorithm rule or integration changes.
- Issue/risk log covering technology, data, adoption and field-process risks.

Client responsibilities

- Provide current Excel schedule files, operational reports, asset master data and available Spinergie/AIS exports.
- Nominate ABL process owners for fleet, port, CTS, cargo/blending, IT and management reporting.
- Facilitate structured discussions with Berau Coal counterparts where required.
- Validate master data: fleet specs, route profiles, jetty constraints, loading rates, CTS capacities, tide/bridge rules and delay codes.
- Participate in UAT, adoption planning and operating-process sign-off.

DECISION PATH

11. Proposed Next Steps

- Conduct a proposal review meeting with ABL to confirm scope, phasing and expected business outcomes.
- Agree whether the engagement starts with a standalone 4-6 week blueprint or proceeds directly as a full implementation with blueprint included.
- Collect sample operating data: OGV programme, jetty plan, tug/barge list, route times, tide/bridge data, Spinergie/AIS exports and delay logs.
- Run a diagnostic workshop with ABL operations, port/blending, CTS, IT and Berau-facing stakeholders.
- Finalize commercial terms, project team, governance cadence and kick-off date.

REFERENCE

Appendix A: Data and Integration Checklist

Data area	Required data elements
Demand	OGV schedule, laycan, ETA/ETB, buyer, quantity, coal quality, layering/hatch plan, priority.
Cargo / Quality	Stockpile source, blend recipe, CV, ash, moisture, sulphur, sample status, release/block status, rain impact.
Port / Jetty	Jetty availability, BLC capacity, loading rate, loading queue, maintenance downtime, operating windows.
Fleet	Owned/chartered status, tug HP, barge size, draft, current location, maintenance, readiness, crew availability.
Route / Environment	Route distance, typical cycle time, tide windows, bridge windows, draft restrictions, weather, safe mooring points.
CTS / OGV	CTS type/capacity, conveyor vs crane/grab, sampler, metal detector, blending capability, OGV hatch/loading progress.
Survey / Certification	Draft survey, hold inspection, sampling, weight certificate, quality certificate, statement of facts.
Actuals / Events	AIS position, geofence events, loading start/end, discharge start/end, waiting reason, breakdown, reassignment, delay reason.

REFERENCE

Appendix B: Key Assumptions Requiring Validation

Assumption area	Validation required during diagnostic
Operating ownership	Confirm ABL and Berau decision rights for demand changes, source changes, delay attribution and recovery approval.
Data quality	Confirm completeness, latency and ownership of Excel schedules, AIS exports, master data, environmental inputs and field event logs.
Integration access	Confirm API/export availability for Spinergie/AIS, environmental constraint inputs, OGV programme inputs and any operational database.
Adoption readiness	Confirm field users, reporting discipline, mobile/web access needs, language requirements and training approach.
Scope boundary	Confirm which Year 2 optimisation and optional external stakeholder-view items are included now versus parked for future phase approval.